Bloch's Proof for Torsion 0-Cycles

Notas de exposición IMATE

César Galindo

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Abstract

These notes rework Bloch's 1979 argument with the center of gravity placed on two intertwined themes: first, the construction of Bloch's cycle map

$$\lambda_X^n \colon \mathrm{CH}^n(X)(\ell) \longrightarrow H^{2n-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(n)),$$

and second, the proof that for $n = \dim X$ this map identifies torsion zero-cycles with torsion in the Albanese variety away from the characteristic. The exposition keeps the geometric focus of the original paper while expanding a number of compressed steps: the Bloch–Ogus input, the Gersten comparison and the role of the Merkurjev–Suslin theorem, the Milnor-K_2 and Tate-symbol description of the first nontrivial Gersten differential, the snake-lemma construction of λ^n, the specialization argument proving the torsionness of the auxiliary map ρ, the reductions to surfaces and generalized Jacobians, the passage through a Lefschetz pencil, the relative Picard torsor, vanishing cycles, and the Picard–Lefschetz formula. Throughout, the goal is not merely to quote the structure of Bloch's proof, but to make transparent why the cycle-theoretic and cohomological constructions fit together.

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1 Bloch's theorem and the structure of the argument

Let k be an algebraically closed field and let X be a smooth projective variety over k. Write

$$A_0(X) := \mathrm{Ker}\big(\deg\colon \mathrm{CH}_0(X) \longrightarrow \mathbb{Z}\big).$$

The Albanese variety of X will be denoted by $\mathrm{Alb}(X)$, and the Albanese map by

$$\mathrm{alb}_X\colon A_0(X) \longrightarrow \mathrm{Alb}(X)(k).$$

For any abelian group M, we write M_{tors} for its torsion subgroup and $M[\ell]$ or $M[\ell^\nu]$ for the kernel of multiplication by ℓ or ℓ^ν.

The result established in detail by Bloch is the following.

Theorem 1.1 (Bloch). *Let X be a smooth projective variety over an algebraically closed field k. Then the Albanese map*

$$\mathrm{alb}_X \colon A_0(X)_{\mathrm{tors}} \longrightarrow \mathrm{Alb}(X)(k)_{\mathrm{tors}}$$

is surjective. Moreover, for every prime $\ell \neq \mathrm{char}(k)$, the induced map on ℓ-primary torsion is an isomorphism. Equivalently,

$$A_0(X)[\ell^\infty] \xrightarrow{\sim} \mathrm{Alb}(X)(k)[\ell^\infty] \qquad (\ell \neq \mathrm{char}(k)).$$

The proof has two very different halves, and it is worth keeping both scales in mind from the start. At the global scale, Bloch first manufactures a cohomological invariant λ^n for torsion cycles of arbitrary codimension. At the geometric scale relevant for 0-cycles, he then proves that in top codimension this invariant is exactly the Albanese-theoretic one.

1. The *surjectivity* is comparatively soft. One uses the divisibility of $A_0(X)$ together with induction on $\dim X$ and weak Lefschetz for hyperplane sections.
2. The *injectivity away from the characteristic* is much subtler. Starting from an ℓ-torsion zero-cycle in the kernel of the Albanese map, Bloch performs a sequence of reductions:
   (i) one replaces the rational-equivalence datum by a divisor on a union of smooth curves with simple combinatorics;
   (ii) one cuts down to a surface;
   (iii) one rewrites the problem inside a generalized Jacobian;
   (iv) one moves the cycle through a Lefschetz pencil to a smooth member;
   (v) one proves a key lemma identifying the kernels of two maps out of the Jacobian of a smooth hyperplane section.

The final step depends on vanishing cycles and the Picard–Lefschetz formula. In particular, Bloch's proof is not merely an argument about Chow groups: it is an argument in which the geometry of Lefschetz pencils controls rational equivalence of zero-cycles.

2 Preliminaries on zero-cycles and Albanese maps

2.1 A presentation of $\mathrm{CH}_0(X)$

Let X be a smooth projective variety. There is a standard presentation

$$\bigoplus_{x \in X_1} k(x)^\times \xrightarrow{\partial} \bigoplus_{x \in X_0} \mathbb{Z} \cdot [x] \longrightarrow \mathrm{CH}_0(X) \longrightarrow 0, \tag{1}$$

where X_i denotes the set of points of X whose Zariski closure has dimension i, and where ∂ is the divisor map along the normalizations of the irreducible curves in X.

Concretely, if $C \subset X$ is an integral curve and $f \in k(C)^\times$, then

$$\partial(f) = \sum_{x \in C^{(1)}} \mathrm{ord}_x(f)[x].$$

Thus rational equivalence on 0-cycles is generated by principal divisors on curves lying on X.

2.2 The Albanese map

Fix a base point $x_0 \in X(k)$. The universal property of the Albanese variety yields a morphism

$$a_X \colon X \longrightarrow \mathrm{Alb}(X)$$

sending x_0 to 0, and hence a homomorphism

$$\mathrm{alb}_X \colon A_0(X) \longrightarrow \mathrm{Alb}(X)(k), \qquad \sum_i n_i[x_i] \longmapsto \sum_i n_i a_X(x_i).$$

The degree-zero condition is exactly what makes the sum independent of the choice of base point.

Two facts will be used repeatedly.

Proposition 2.1. *Let $\pi \colon \widetilde{X} \longrightarrow X$ be the blow-up of a smooth projective variety along a smooth center. Then*

$$\pi_* \colon \mathrm{CH}_0(\widetilde{X}) \xrightarrow{\sim} \mathrm{CH}_0(X) \qquad \textit{and} \qquad \mathrm{Alb}(\widetilde{X}) \xrightarrow{\sim} \mathrm{Alb}(X)$$

are isomorphisms.

Proof. For CH_0 one uses the localization sequence and the fact that the exceptional fibres are rational varieties; in particular every degree-zero 0-cycle supported on an exceptional fibre is rationally equivalent to zero. For the Albanese, one uses birational invariance of $H^1(X, \mathcal{O}_X)$ and of the Picard variety for smooth projective varieties, hence of the dual abelian variety. □

2.3 Generalized Jacobians of singular curves

Let C be a reduced projective curve over k. We denote by $J(C)$ its generalized Jacobian. For the argument below only the following properties are needed.

- There is a natural degree-zero Abel map from smooth points of C to $J(C)$, and every degree-zero zero-cycle supported on C_{sm} defines a point of $J(C)$.
- If $\nu \colon \widetilde{C} \longrightarrow C$ is the normalization, then the kernel of $J(C) \longrightarrow J(\widetilde{C})$ is a connected linear algebraic group; in particular it is divisible on k-points.
- If $C = \bigcup_i C_i$ has only ordinary double points and if no support point of a divisor lies at an intersection $C_i \cap C_j$, then degree-zero divisors on the C_i define classes in $J(C)$ exactly as one expects from the smooth case.

These facts are the formal device allowing Bloch to convert a rational-equivalence relation

$$\ell z = \partial(f_1, \ldots, f_r)$$

into the statement that the class of ℓz vanishes in a suitable generalized Jacobian.

3 Bloch's construction of the cycle map λ^n

Before entering the proof for 0-cycles, it is worth pausing over the construction that occupies Bloch's §2. This is the real conceptual hinge of the paper. The theorem on torsion zero-cycles is not an isolated miracle: it sits at the end of a general cohomological machine which, for every codimension n, attaches to an ℓ-primary torsion cycle a class in degree $2n-1$ étale cohomology. Bloch's point is that rational equivalence is presented by principal divisors

on codimension-$(n-1)$ subvarieties, while the Bloch–Ogus coniveau machine presents the sheaf $\mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})$ by exactly the same codimension filtration. The cycle map λ^n is obtained by making these two presentations talk to each other.

For the rest of this section fix an integer $n \geq 1$ and a prime $\ell \neq \mathrm{char}(k)$. We write

$$\mathrm{CH}^n(X)(\ell) := \varinjlim_{\nu} \mathrm{CH}^n(X)[\ell^\nu]$$

for the ℓ-primary torsion subgroup.

3.1 The two presentations one wants to compare

There are two a priori different ways to encode codimension-n cycles.

(1) *The cycle-theoretic presentation.* Rational equivalence is generated by principal divisors on integral codimension-$(n-1)$ subvarieties. Concretely, one has an exact sequence

$$\bigoplus_{x \in X^{n-1}} k(x)^* \xrightarrow{\delta} \bigoplus_{x \in X^n} \mathbb{Z} \longrightarrow \mathrm{CH}^n(X) \longrightarrow 0. \tag{2}$$

The map δ sends a rational function on a codimension-$(n-1)$ point to its divisor along the codimension-n points lying on its closure.

(2) *The coniveau-cohomological presentation.* The Bloch–Ogus theorem identifies the E_2-page of the coniveau spectral sequence with the Zariski cohomology of the sheaves $\mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})$. In the degree relevant for codimension n, Corollary 1.5 of Bloch says that

$$H^{n-1}\big(X, \mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})\big) \cong H^{n-1}\Big[\bigoplus_{x \in X^{n-2}} H^2\big(k(x), \mu_{\ell^\nu}^{\otimes 2}\big) \longrightarrow \bigoplus_{x \in X^{n-1}} k(x)^*/k(x)^{*\ell^\nu} \xrightarrow{\delta_\nu} \bigoplus_{x \in X^n} \mathbb{Z}/\ell^\nu\mathbb{Z}\Big]. \tag{3}$$

The differential δ_ν is simply the reduction of δ modulo ℓ^ν.

The strategy is now visible. A torsion cycle $z \in \mathrm{CH}^n(X)[\ell^\nu]$ is represented by a codimension-n cycle whose multiple by ℓ^ν is rationally equivalent to zero. This means that, after choosing rational functions on codimension-$(n-1)$ subvarieties, one obtains an element in the kernel of δ_ν. If that kernel can be sent functorially to

$$H^{2n-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes n}),$$

then one may hope to attach a cohomology class to z. The whole construction consists in making this naive idea canonical.

3.2 Bloch–Ogus input in the precise range needed

For integers q, ν, n, let

$$\mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})$$

denote the Zariski sheaf on X associated to the presheaf

$$U \longmapsto H^q_{\acute{e}t}(U, \mu_{\ell^\nu}^{\otimes n}).$$

Bloch starts from the Leray spectral sequence for the morphism of sites $X_{\acute{e}t} \longrightarrow X_{\mathrm{Zar}}$,

$$E_2^{p,q} = H^p\big(X, \mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})\big) \Longrightarrow H^{p+q}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes n}), \tag{4}$$

and compares it with the coniveau spectral sequence

$$E_1^{p,q} = \bigoplus_{x \in X^p} H^{q-p}\big(k(x), \mu_{\ell^\nu}^{\otimes(n-p)}\big) \Longrightarrow H_{\acute{e}t}^{p+q}(X, \mu_{\ell^\nu}^{\otimes n}). \tag{5}$$

The Bloch–Ogus theorem identifies these spectral sequences from E_2 onward, and the localized E_1-complex gives an acyclic resolution of the sheaf $\mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})$; for the underlying coniveau formalism see Bloch–Ogus [2], while Bloch's use of it in the present context is concentrated in [1, § 1–2]. In the range relevant for codimension n cycles this yields two concrete consequences.

Corollary 3.1 (Bloch–Ogus vanishing and boundary)**.** *For every $\nu \geq 1$ one has:*

(i) the vanishing

$$H^p\big(X, \mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})\big) = 0 \qquad (p > q);$$

(ii) since $E_2^{n+i,n-i+1} = 0$ for $i \geq 1$, the Leray spectral sequence yields a natural boundary map

$$\beta_\nu \colon H^{n-1}\big(X, \mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})\big) \longrightarrow H_{\acute{e}t}^{2n-1}(X, \mu_{\ell^\nu}^{\otimes n}). \tag{6}$$

The first point says that the troublesome higher Zariski cohomology groups above the diagonal disappear. The second point says that the class produced on the E_2-page has a canonical image in total degree $2n-1$. This is the cohomological half of the construction.

3.3 The snake-lemma diagram

The cycle-theoretic half is encoded in the diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \bigoplus_{x \in X^{n-1}} k(x)^*/k^* \arrow[r, "\ell^\nu"] \arrow[d, "\delta"] & \bigoplus_{x \in X^{n-1}} k(x)^*/k^* \arrow[r] \arrow[d, "\delta"] & \bigoplus_{x \in X^{n-1}} k(x)^*/k(x)^{*\ell^\nu} \arrow[r] \arrow[d, "\delta_\nu"] & 0 \\
0 \arrow[r] & \bigoplus_{x \in X^n} \mathbb{Z} \arrow[r, "\ell^\nu"] & \bigoplus_{x \in X^n} \mathbb{Z} \arrow[r] & \bigoplus_{x \in X^n} \mathbb{Z}/\ell^\nu\mathbb{Z} \arrow[r] & 0.
\end{tikzcd} \tag{7}$$

Here $k^* \subset k(x)^*$ denotes the subgroup of constant functions. Quotienting by constants is exactly what one needs in order to make the specialization argument later behave well: constants are ℓ-divisible over an algebraically closed field, so they do not contribute to divisors in codimension n.

Applying the snake lemma to (7), one gets an exact sequence

$$0 \longrightarrow \mathrm{Ker}(\delta)/\ell^\nu \,\mathrm{Ker}(\delta) \longrightarrow \mathrm{Ker}(\delta_\nu) \longrightarrow \mathrm{CH}^n(X)[\ell^\nu] \longrightarrow 0. \tag{8}$$

This exact row is the algebraic heart of the construction, and it deserves to be read slowly.

- An element of $\mathrm{Ker}(\delta_\nu)$ is represented by a family of rational functions on codimension-$(n-1)$ subvarieties whose divisor becomes zero modulo ℓ^ν.
- The map to $\mathrm{CH}^n(X)[\ell^\nu]$ sends such a family to the corresponding codimension-n cycle class; exactness says precisely that this cycle class is ℓ^ν-torsion.
- Two lifts of the same torsion cycle differ by an honest element of $\mathrm{Ker}(\delta)$, modulo ℓ^ν.

Thus every torsion cycle can be lifted to a coniveau cocycle, but not canonically. The remaining problem is to show that the ambiguity becomes harmless in $\mathbb{Q}_\ell/\mathbb{Z}_\ell$-cohomology.

3.4 From $\mathrm{Ker}(\delta_\nu)$ to étale cohomology

Because δ_ν is the last differential in the complex (3), an element of $\mathrm{Ker}(\delta_\nu)$ determines a cohomology class in

$$H^{n-1}\big(X, \mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})\big).$$

We denote the resulting map by

$$\alpha_\nu\colon\ \mathrm{Ker}(\delta_\nu) \longrightarrow H^{n-1}\big(X, \mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})\big). \tag{9}$$

Composing with the boundary map (6), one obtains

$$\beta_\nu \circ \alpha_\nu\colon\ \mathrm{Ker}(\delta_\nu) \longrightarrow H^{2n-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes n}). \tag{10}$$

Bloch packages this together with (8) in the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Ker}(\delta)/\ell^\nu\,\mathrm{Ker}(\delta) \arrow[r] \arrow[ddr, bend right, "\rho_\nu"'] & \mathrm{Ker}(\delta_\nu) \arrow[r] \arrow[d, "\alpha_\nu"] & \mathrm{CH}^n(X)[\ell^\nu] \arrow[r] & 0 \\
& & H^{n-1}\big(X, \mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})\big) \arrow[d, "\beta_\nu"] & & \\
& & H^{2n-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes n}) & &
\end{tikzcd} \tag{11}$$

where ρ_ν is the restriction of $\beta_\nu \circ \alpha_\nu$ to the left-hand term. Conceptually, ρ_ν measures what happens when the chosen lift of a torsion cycle is changed by a genuine rational-equivalence datum.

3.5 Passing to the inverse system and the obstruction map ρ

The transition maps in $\mu_{\ell^\nu}^{\otimes n}$ are compatible with all constructions above, so the maps ρ_ν assemble into a morphism

$$\rho\colon\ \mathrm{Ker}(\delta) \longrightarrow H^{2n-1}_{\acute{e}t}(X, \mathbb{Z}_\ell(n)). \tag{12}$$

This is Bloch's equation (2.3). The whole issue is now concentrated in the following statement.

Lemma 3.2 (Bloch, Lemma 2.4)**.** *The image of ρ is torsion.*

Expanded proof. The argument has two distinct stages.

Step 1: the case of finite-field origin. Assume first that $k = \overline{\mathbb{F}}_p$ and that both X and a chosen element $x \in \mathrm{Ker}(\delta)$ descend to a finite field $\mathbb{F}_{p^a}$. Let

$$f = (p^a\text{-power Frobenius on } X_0) \times (\mathrm{id}_{\overline{\mathbb{F}}_p}).$$

Because the construction of ρ is functorial, Frobenius acts compatibly and one has

$$f\rho(x) = \rho(fx) = p^{an}\rho(x).$$

The factor p^{an} appears because the Tate twist (n) compensates for weight $2n$. By Deligne's proof of the Weil conjectures, p^{an} is not an eigenvalue of Frobenius on

$$H^{2n-1}_{ét}(X, \mathbb{Q}_\ell(n)).$$

Therefore the equality above can hold only if $\rho(x)$ is torsion.

Step 2: reduction from an arbitrary algebraically closed field. For general algebraically closed k, Bloch spreads X out to a smooth scheme $\mathscr{X}$ over the spectrum of a valuation ring R with residue field $k_0 = \overline{\mathbb{F}}_p$. Cospecialization yields an isomorphism

$$s\colon H^{2n-1}_{ét}(X_k, \mathbb{Z}_\ell(n)) \xrightarrow{\sim} H^{2n-1}_{ét}(X_0, \mathbb{Z}_\ell(n)). \tag{13}$$

At the level of the coniveau complexes, local acyclicity gives specialization maps on local cohomology, hence a morphism of E_1-complexes. On the cycle-theoretic side, one specializes closures of codimension-$(n-1)$ and codimension-n points. Because the constant subgroups

$$\bigoplus_{x \in X^{n-1}} k^* \subset \bigoplus_{x \in X^{n-1}} k(x)^*$$

and

$$\bigoplus_{x \in X_0^{n-1}} k_0^* \subset \bigoplus_{x \in X_0^{n-1}} k_0(x)^*$$

are ℓ-divisible and map trivially under divisor maps, one gets a commutative specialization diagram with exact rows

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{Ker}(\overline{\delta}_k) \arrow[r] \arrow[d, "t"] & \bigoplus_{x \in X_k^{n-1}} k(x)^*/k^* \arrow[r] \arrow[d, "t"] & \bigoplus_{x \in X_k^{n}} \mathbb{Z} \arrow[r] \arrow[d, "t"] & \mathrm{CH}^n(X_k) \arrow[r] \arrow[d, "t"] & 0 \\
0 \arrow[r] & \mathrm{Ker}(\overline{\delta}_0) \arrow[r] & \bigoplus_{x \in X_0^{n-1}} k_0(x)^*/k_0^* \arrow[r] & \bigoplus_{x \in X_0^{n}} \mathbb{Z} \arrow[r] & \mathrm{CH}^n(X_0) \arrow[r] & 0.
\end{tikzcd}$$

Bloch checks that ρ factors through these specialized kernels and that

$$\rho_0 \circ t = s \circ \rho_k.$$

Since the image of ρ_0 is torsion by Step 1 and s is an isomorphism, the image of ρ_k is torsion as well. This finishes the proof. □

This lemma is the decisive point of the whole construction. Up to this moment one only knows that different lifts of a torsion cycle may differ by an element of $\mathrm{Ker}(\delta)$, and hence may change the cohomology class by something in the image of ρ. The lemma says that this ambiguity already lands in torsion inside the integral ℓ-adic group. After passing to the divisible quotient $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, the ambiguity disappears.

3.6 The direct limit and the definition of λ^n

Taking the direct limit of (8) over ν gives an exact sequence

$$0 \longrightarrow \mathrm{Ker}(\delta) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \longrightarrow \varinjlim_{\nu} \mathrm{Ker}(\delta_\nu) \longrightarrow \mathrm{CH}^n(X)(\ell) \longrightarrow 0. \tag{14}$$

Likewise, the compatible maps (10) induce a homomorphism

$$\widetilde{\lambda}_X^n \colon \varinjlim_{\nu} \operatorname{Ker}(\delta_\nu) \longrightarrow H^{2n-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(n)). \tag{15}$$

What happens on the left-hand subgroup in (14)? By construction it is exactly the image of

$$\operatorname{Ker}(\delta) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \xrightarrow{\rho\otimes\mathrm{id}} H^{2n-1}_{\acute{e}t}(X, \mathbb{Z}_\ell(n)) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \longrightarrow H^{2n-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(n)).$$

Because $\operatorname{Im}(\rho)$ is torsion by Theorem 3.2, its image in the divisible quotient

$$H^{2n-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(n))$$

is zero. Therefore $\widetilde{\lambda}_X^n$ vanishes on the left-hand term of (14), and hence factors uniquely through the quotient.

Definition 3.3. The *Bloch cycle map* is the induced homomorphism

$$\lambda_X^n \colon \operatorname{CH}^n(X)(\ell) \longrightarrow H^{2n-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(n)). \tag{16}$$

Following Bloch, one inserts an overall minus sign in this definition so that the codimension-one normalization agrees with the usual Kummer map.

In more pedestrian terms, if $z \in \operatorname{CH}^n(X)[\ell^\nu]$, one chooses a lift

$$\widetilde{z}_\nu \in \operatorname{Ker}(\delta_\nu)$$

of z through (8), applies $\beta_\nu \circ \alpha_\nu$, and then views the result in

$$H^{2n-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(n)).$$

A different lift changes the answer by the image of an element of $\operatorname{Ker}(\delta)$, hence by an element of the image of ρ, and this vanishes after passage to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$. That is the complete reason why the definition is independent of choices.

3.7 A commutative summary diagram

The whole construction may be summarized by the commutative pattern

$$\begin{tikzcd}
\operatorname{Ker}(\delta_\nu) \arrow[r, "\alpha_\nu"] \arrow[d] & H^{n-1}\big(X, \mathcal{H}^n(\mu_{\ell^\nu}^{\otimes n})\big) \arrow[d, "\beta_\nu"] \\
\operatorname{CH}^n(X)[\ell^\nu] \arrow[r, dashed] & H^{2n-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes n}),
\end{tikzcd}$$

where the lower horizontal arrow is only partially defined at the finite level because one has not yet killed the ambiguity coming from $\operatorname{Ker}(\delta)$. Passing to the direct limit and quotienting by that ambiguity produces the genuine map λ_X^n.

This is the conceptual architecture of Bloch's §2: the left vertical arrow is supplied by cycle theory, the top horizontal arrow by the coniveau resolution, the right vertical arrow by the Leray spectral sequence, and Lemma 2.4 is the bridge that allows everything to descend from lifts to Chow classes.

3.8 The codimension-one normalization

Bloch checks in Proposition 3.6 that when $n = 1$, the map λ^1_X is exactly the Kummer boundary map for line bundles. Since this normalization fixes the sign convention, it is worth spelling it out.

For $n = 1$, the presentation (2) becomes the standard divisor sequence

$$k(X)^* \xrightarrow{\mathrm{div}} \bigoplus_{x \in X^1} \mathbb{Z} \longrightarrow \mathrm{Pic}(X) \longrightarrow 0.$$

Modulo ℓ^ν, one gets

$$0 \longrightarrow \mathcal{O}_X^*/\mathcal{O}_X^{*\ell^\nu} \longrightarrow (k(X)^*/k(X)^{*\ell^\nu})_X \xrightarrow{\mathrm{div}_\nu} \bigoplus_{x \in X^1} i_{x,*}\mathbb{Z}/\ell^\nu\mathbb{Z} \longrightarrow 0,$$

and therefore a commutative cohomology diagram whose left vertical arrow is the Kummer boundary. If $f \in k(X)^*$ represents an element of $\mathrm{Ker}(\mathrm{div}_\nu)$, choose an open cover $\{U_\alpha\}$ on which $f = g_\alpha^{\ell^\nu}$. Then on overlaps one obtains a μ_{ℓ^ν}-cocycle

$$g_\alpha g_\beta^{-1} \in \Gamma(U_\alpha \cap U_\beta, \mu_{\ell^\nu}),$$

which is exactly the Kummer class of the line bundle associated with the divisor $-\ell^{-\nu}\,\mathrm{div}(f)$. The minus sign is precisely the one Bloch inserts in the definition of λ so that

$$\lambda^1_X\colon\ \mathrm{Pic}(X)(\ell) \xrightarrow{\sim} H^1_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$$

coincides with the usual Kummer isomorphism.

3.9 Why this matters for the theorem on 0-cycles

The general construction above specializes in two extremal ways emphasized by Bloch.

(i) For $n = 1$, λ^1_X recovers the Kummer description of torsion line bundles.

(ii) For $n = d = \dim X$, the map

$$\lambda^d_X\colon\ \mathrm{CH}_0(X)(\ell) \longrightarrow H^{2d-1}_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d))$$

is the cohomological shadow of the torsion Albanese map. The long geometric argument in §4 of Bloch is exactly what identifies this abstractly constructed cohomological invariant with the concrete algebro-geometric one carried by $\mathrm{Alb}(X)$.

So the proof for 0-cycles should be read in the following order. First one constructs λ^d by a general coniveau argument. Then one proves, by a sequence of geometric reductions and finally by Lefschetz-pencil monodromy, that in top codimension the classes detected by λ^d are exactly those detected by the Albanese variety. The theorem is therefore both a statement about zero-cycles and a statement about the reach of the coniveau description of étale cohomology.

4 The seminar reformulation: Gersten, Merkurjev–Suslin, and the top-codimension bridge

The seminar version of the story is slightly different in emphasis from the route we have followed so far. Instead of first presenting the snake-lemma construction and only later re-entering the top-codimension case, one can pause at this point and reorganize the same material into a three-step bridge:

(1) the Bloch–Ogus resolution identifies the relevant Zariski cohomology groups and supplies the edge map to étale cohomology;

(2) the comparison with the tail of the Gersten complex, together with the Merkurjev–Suslin theorem, isolates the precise quotient in which torsion cycles live;

(3) in top codimension, Poincaré duality identifies the resulting étale cohomology group with ℓ-primary torsion in the Albanese variety.

This is exactly the package emphasized in the seminar notes you attached, and it is worth spelling it out as a coherent interlude before returning to the geometric reductions of Bloch's proof.

4.1 The Bloch–Ogus resolution revisited

For a smooth variety X and integers q, n, ν, the coniveau spectral sequence gives an acyclic resolution of

$$\mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})$$

by skyscraper sheaves supported on points of fixed codimension. Unwinding the localized E_1-page, one gets the complex

$$0 \longrightarrow \mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n}) \longrightarrow \bigoplus_{x\in X^0} i_{x,*}H^q\big(k(x), \mu_{\ell^\nu}^{\otimes n}\big) \longrightarrow \bigoplus_{x\in X^1} i_{x,*}H^{q-1}\big(k(x), \mu_{\ell^\nu}^{\otimes (n-1)}\big) \longrightarrow \cdots, \quad (17)$$

which is acyclic. Consequently,

$$E_2^{p,q} = H^p\big(X, \mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})\big) \qquad \text{and} \qquad H^p\big(X, \mathcal{H}^q(\mu_{\ell^\nu}^{\otimes n})\big) = 0 \quad (p > q).$$

In particular, for codimension p cycles there is a canonical edge map

$$H^{p-1}\big(X, \mathcal{H}^p(\mu_{\ell^\nu}^{\otimes p})\big) \longrightarrow H^{2p-1}_{\acute{e}t}\big(X, \mu_{\ell^\nu}^{\otimes p}\big). \quad (18)$$

This is the same map denoted β_ν in Equation (6), but written now in the specific degree in which codimension-p cycle classes live.

The second standard consequence of Bloch–Ogus is the formula

$$H^p\big(X, \mathcal{H}^p(\mu_{\ell^\nu}^{\otimes p})\big) \cong \mathrm{CH}^p(X)/\ell^\nu. \quad (19)$$

What matters here is not only the isomorphism itself, but also its meaning: the last nonzero differential in the Bloch–Ogus complex already remembers exactly the codimension-p divisor relations, hence exactly the reduction of Chow groups modulo ℓ^ν.

4.2 Comparison with the tail of the Gersten complex

Let $\mathcal{K}_p$ denote the Zariski sheaf associated with Quillen's K_p. The part of the Gersten complex relevant for codimension p is

$$\bigoplus_{x\in X^{p-2}} K_2\big(k(x)\big) \longrightarrow \bigoplus_{x\in X^{p-1}} k(x)^\times \longrightarrow \bigoplus_{x\in X^p} \mathbb{Z} \longrightarrow 0. \quad (20)$$

The left-hand map is built from tame symbols, and the middle map is the divisor map. On the cohomological side, the corresponding tail of the Bloch–Ogus complex is

$$\bigoplus_{x\in X^{p-2}} H^2\big(k(x), \mu_{\ell^\nu}^{\otimes 2}\big) \longrightarrow \bigoplus_{x\in X^{p-1}} H^1\big(k(x), \mu_{\ell^\nu}\big) \longrightarrow \bigoplus_{x\in X^p} H^0\big(k(x), \mathbb{Z}/\ell^\nu\mathbb{Z}\big) \longrightarrow 0. \quad (21)$$

The bridge between Equation (20) and Equation (21) is Kummer theory in degree 1 and the Merkurjev–Suslin theorem in degree 2.

Milnor K_2, Steinberg symbols, and the tame boundary. For a field F, Milnor's group $K_2^M(F)$ is defined by

$$K_2^M(F) := F^\times \otimes_{\mathbb{Z}} F^\times \Big/ \langle a \otimes (1-a)\ ;\ a \neq 0, 1\rangle,$$

with class of $a \otimes b$ denoted by $\{a, b\}$. By Matsumoto's theorem one has a canonical isomorphism

$$K_2^M(F) \xrightarrow{\sim} K_2(F),$$

so in the field-valued range relevant for the tail of the Gersten complex one may pass freely between Milnor symbols and Quillen's K_2; see Milnor [4, Chapter IV] and Quillen [7]. If v is a discrete valuation on F with valuation ring A_v, residue field $\kappa(v)$, and uniformizer π, the tame symbol is the homomorphism

$$\partial_v \colon K_2^M(F) \longrightarrow \kappa(v)^\times, \qquad \partial_v\{a, b\} = (-1)^{v(a)v(b)}\, \overline{a^{v(b)} b^{-v(a)}}.$$

This is the first genuinely higher-K-theoretic term in the codimension filtration: the map

$$\bigoplus_{x \in X^{p-2}} K_2\big(k(x)\big) \longrightarrow \bigoplus_{x \in X^{p-1}} k(x)^\times$$

is built exactly from these tame symbols, and the next differential is the usual divisor map. In other words, the passage from codimension $p-1$ to codimension p is still controlled by rational functions, but the previous step already remembers symbols in K_2.

There are two points worth making explicit here. First, Milnor K_2 is not introduced for ornament: it is the unique place where one can simultaneously remember two rational functions before taking divisors. Second, Quillen's localization sequence globalizes these local tame symbols into the Gersten resolution, so that in the regular case one really has the tail

$$0 \longrightarrow \mathcal{K}_p \longrightarrow K_p(k(X)) \longrightarrow \bigoplus_{x \in X^1} K_{p-1}\big(k(x)\big) \longrightarrow \cdots \longrightarrow \bigoplus_{x \in X^p} K_0\big(k(x)\big) \longrightarrow 0,$$

exact in the Zariski topology for the regular schemes considered by Bloch and Bloch–Ogus. In top codimension, this is the precise reason the comparison diagram starts one step earlier than divisors: the codimension-$p-2$ term is already measuring the failure of two functions to commute under specialization.

4.3 Local symbols, residues, and why the K_2-term is exactly right

At this point it is worth pausing over the meaning of the leftmost term in the Gersten strip. The differential

$$\bigoplus_{x \in X^{p-2}} K_2\big(k(x)\big) \longrightarrow \bigoplus_{y \in X^{p-1}} k(y)^\times$$

is not an incidental correction term: it is the first place where codimension-two incidence data enters the cycle-theoretic picture. If $x \in X^{p-2}$ and $\overline{\{x\}}$ is the corresponding integral subvariety, then each codimension-one point $y \in \overline{\{x\}}^{(1)}$ defines a discrete valuation on $k(x)$, and the component of the Gersten differential from x to y is exactly the tame symbol for that valuation. Thus the whole codimension-two to codimension-one passage is controlled by local symbols.

From this viewpoint, Bloch's use of Bloch–Ogus and Merkurjev–Suslin is extremely natural. The codimension-one to codimension-p part of the cycle calculus is governed

by divisors of rational functions. The step before that must therefore remember how two rational functions interact along a codimension-one valuation. Milnor's K_2 is precisely the universal receptacle for that interaction, and the tame symbol is the operation that extracts the codimension-one contribution of a K_2-class.

The Tate tame symbol and the first residue square. Following Tate's reinterpretation of the norm-residue formalism [9], one may regard the tame symbol as the local boundary underlying the cup-product symbol in Galois cohomology. Kummer theory gives

$$h^1_{\ell^\nu,F}\colon F^\times/F^{\times\ell^\nu} \xrightarrow{\sim} H^1\big(F,\mu_{\ell^\nu}\big),$$

and cupping two degree-one classes yields the Galois symbol

$$h^2_{\ell^\nu,F}\colon K_2^M(F)/\ell^\nu \longrightarrow H^2\big(F,\mu_{\ell^\nu}^{\otimes 2}\big).$$

For a discrete valuation v of F with residue field $\kappa(v)$, the cohomological residue

$$\partial_v^{\acute{e}t}\colon H^2\big(F,\mu_{\ell^\nu}^{\otimes 2}\big) \longrightarrow H^1\big(\kappa(v),\mu_{\ell^\nu}\big)$$

corresponds, under Kummer theory, exactly to the tame symbol

$$\partial_v\colon K_2^M(F)/\ell^\nu \longrightarrow \kappa(v)^\times/\kappa(v)^{\times\ell^\nu}.$$

The comparison is encoded by the commutative square

$$\begin{tikzcd}
K_2^M(F)/\ell^\nu \arrow[r, "h^2_{\ell^\nu,F}", "\sim"'] \arrow[d, "\partial_v^{\mathrm{tame}}"] & H^2\big(F,\mu_{\ell^\nu}^{\otimes 2}\big) \arrow[d, "\partial_v^{\acute{e}t}"] \\
\kappa(v)^\times/\kappa(v)^{\times\ell^\nu} \arrow[r, "h^1_{\ell^\nu,\kappa(v)}", "\sim"'] & H^1\big(\kappa(v),\mu_{\ell^\nu}\big).
\end{tikzcd}$$

In Bloch's proof this square is the hinge between the Gersten presentation of Chow groups and the coniveau presentation of étale cohomology.

Two successive residues and the codimension-two to codimension-p transition. Because the next differential in both complexes is the divisor map, one really has a two-step compatibility. If w is a discrete valuation of $\kappa(v)$, then the valuation map modulo ℓ^ν corresponds to the cohomological residue in degree 1, and one gets

$$\begin{tikzcd}
K_2^M(F)/\ell^\nu \arrow[r, "h^2"] \arrow[d, "\partial_v^{\mathrm{tame}}"] & H^2\big(F,\mu_{\ell^\nu}^{\otimes 2}\big) \arrow[d, "\partial_v^{\acute{e}t}"] \\
\kappa(v)^\times/\kappa(v)^{\times\ell^\nu} \arrow[r, "h^1", "\sim"'] \arrow[d, "\mathrm{ord}_w \bmod \ell^\nu"] & H^1\big(\kappa(v),\mu_{\ell^\nu}\big) \arrow[d, "\partial_w^{\acute{e}t}"] \\
\mathbb{Z}/\ell^\nu\mathbb{Z} \arrow[r, equal] & \mathbb{Z}/\ell^\nu\mathbb{Z}.
\end{tikzcd}$$

This is the local model for the last two arrows in the Gersten and Bloch–Ogus tails. Once one sheafifies and sums over points of codimension $p-2$, $p-1$, and p, one gets the global comparison diagram used to manufacture the exact sequence whose quotient is $\mathrm{CH}^p(X)[\ell^\nu]$.

This is also the point where the norm-residue philosophy enters in exactly the strength needed by Bloch's argument. One does not need the full Bloch–Kato theorem in all degrees. One needs precisely the statement that degree-two Galois cohomology with $\mu_{\ell^\nu}^{\otimes 2}$ coefficients is exhausted by Milnor symbols modulo ℓ^ν, compatibly with residues. Merkurjev–Suslin is therefore not an external embellishment but the exact theorem that turns the first nontrivial higher-K-theoretic residue into the first nontrivial cohomological residue.

Why this matters later for Picard–Lefschetz theory. The compatibility just described should be remembered when the proof turns to Lefschetz pencils. On the K-theoretic/cohomological side one studies boundary maps attached to codimension-one valuations; on the geometric side of the pencil one studies variation maps attached to codimension-one degenerations of smooth fibres. These are not the same maps, but they play analogous structural roles: both measure what is created when one crosses a divisor of singularities. Bloch's proof works because the cycle map packages the first kind of boundary calculus, while the key lemma is proved by controlling the second kind via vanishing cycles and Picard–Lefschetz reflections.

The Galois symbol and why Merkurjev–Suslin is the precise comparison theorem. The theorem of Merkurjev and Suslin says that the Galois symbol is an isomorphism for every field F and every prime $\ell \neq \mathrm{char}(F)$; see [5]. The point is not merely that both sides have the same size: the symbol is compatible with the tame boundary on the K-theory side and the residue map on the cohomological side. Put differently, Milnor K-theory explains the symbols, Quillen K-theory globalizes them into a Gersten resolution, and Merkurjev–Suslin identifies the only genuinely new piece with degree-two Galois cohomology.

Theorem 4.1 (Merkurjev–Suslin)**.** *For every field F and every prime $\ell \neq \mathrm{char}(F)$, the Galois symbol induces an isomorphism*

$$K_2(F)/\ell^\nu K_2(F) \xrightarrow{\sim} H^2\big(F, \mu_{\ell^\nu}^{\otimes 2}\big).$$

Moreover, if F is the fraction field of a discrete valuation ring with residue field κ, the tame symbol and the residue map fit into a commutative square

$$\begin{tikzcd} K_2(F) \arrow[r] \arrow[d, "T"] & H^2\big(F, \mu_{\ell^\nu}^{\otimes 2}\big) \arrow[d] \\ \kappa^\times \arrow[r] & H^1\big(\kappa, \mu_{\ell^\nu}\big). \end{tikzcd} \tag{22}$$

For the seminar discussion, the theorem should be read together with Milnor's description of symbols, the Quillen–Gersten resolution, and the residue compatibility above; this is exactly the package used implicitly in Bloch's [1, § 2]. This theorem is precisely what makes the seminar comparison work: the left vertical arrows in the Gersten strip become exact after reduction modulo ℓ^ν, while the middle vertical arrows are exact by Kummer theory. Consequently one obtains the following diagram with exact rows and exact columns in the

first two vertical strips:

$$\begin{tikzcd}
\bigoplus_{x\in X^{p-2}} K_2(k(x)) \arrow[r] \arrow[d] & \bigoplus_{x\in X^{p-1}} k(x)^\times \arrow[r] \arrow[d] & \bigoplus_{x\in X^{p}} \mathbb{Z} \arrow[r] \arrow[d] & 0 \\
\bigoplus_{x\in X^{p-2}} K_2(k(x)) \arrow[r] \arrow[d] & \bigoplus_{x\in X^{p-1}} k(x)^\times \arrow[r] \arrow[d] & \bigoplus_{x\in X^{p}} \mathbb{Z} \arrow[r] \arrow[d] & 0 \\
\bigoplus_{x\in X^{p-2}} H^2(k(x),\mu_{\ell^\nu}^{\otimes 2}) \arrow[r] & \bigoplus_{x\in X^{p-1}} H^1(k(x),\mu_{\ell^\nu}) \arrow[r] & \bigoplus_{x\in X^{p}} H^0(k(x),\mathbb{Z}/\ell^\nu\mathbb{Z}) \arrow[r] & 0.
\end{tikzcd} \tag{23}$$

The upper two rows are the last three terms of the Gersten resolution for $\mathcal{K}_p$, while the bottom row is the corresponding part of the Bloch–Ogus complex. The point of Equation (23) is that the quotient between these two resolutions is measured exactly by ℓ^ν-torsion in Chow groups.

Proposition 4.2. *For every $p \geq 1$ and every $\nu \geq 1$, there is a natural short exact sequence*

$$0 \longrightarrow H^{p-1}(X,\mathcal{K}_p)/\ell^\nu \longrightarrow H^{p-1}\big(X,\mathcal{H}^p(\mu_{\ell^\nu}^{\otimes p})\big) \longrightarrow \mathrm{CH}^p(X)[\ell^\nu] \longrightarrow 0. \tag{24}$$

Passing to the direct limit over ν gives

$$0 \longrightarrow H^{p-1}(X,\mathcal{K}_p)\otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \longrightarrow H^{p-1}\big(X,\mathcal{H}^p(\mathbb{Q}_\ell/\mathbb{Z}_\ell(p))\big) \longrightarrow \mathrm{CH}^p(X)(\ell) \longrightarrow 0. \tag{25}$$

Proof. Take cohomology of the diagram Equation (23). The upper two rows compute the last piece of the Gersten resolution of $\mathcal{K}_p$ for a smooth scheme; on this exactness, see Quillen [7] together with the way Bloch–Ogus splice the Gersten and coniveau formalisms in [2]. Hence their cohomology in degree $p-1$ is $H^{p-1}(X,\mathcal{K}_p)/\ell^\nu$. The bottom row computes $H^{p-1}(X,\mathcal{H}^p(\mu_{\ell^\nu}^{\otimes p}))$. Finally, the cokernel of the middle divisor map is, by Equation (19), exactly $\mathrm{CH}^p(X)/\ell^\nu$, and the portion detected by the comparison diagram is its ℓ^ν-torsion subgroup. Chasing kernels and cokernels gives Equation (24); the direct-limit statement follows because $\varinjlim_\nu M/\ell^\nu M \cong M\otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ for any abelian group M. □

4.4 How the exact sequence factors through Bloch's cycle map

The next point is exactly the one that appears slightly mysteriously in seminar-style presentations: why does the edge map to étale cohomology descend from the middle term of Equation (25) to the quotient $\mathrm{CH}^p(X)(\ell)$? In our earlier language, this is nothing but the statement that the ambiguity measured by the obstruction map ρ becomes invisible after passage to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$.

Proposition 4.3. *The edge morphism*

$$H^{p-1}\big(X,\mathcal{H}^p(\mathbb{Q}_\ell/\mathbb{Z}_\ell(p))\big) \longrightarrow H^{2p-1}_{\acute{e}t}\big(X,\mathbb{Q}_\ell/\mathbb{Z}_\ell(p)\big)$$

vanishes on the subgroup $H^{p-1}(X,\mathcal{K}_p)\otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$. Hence it factors uniquely through a morphism

$$\lambda^p_X\colon \mathrm{CH}^p(X)(\ell) \longrightarrow H^{2p-1}_{\acute{e}t}\big(X,\mathbb{Q}_\ell/\mathbb{Z}_\ell(p)\big). \tag{26}$$

Proof. This is the sheaf-theoretic reformulation of the construction already carried out in §2. Indeed, the left-hand subgroup in Equation (25) corresponds to the image of $\mathrm{Ker}(\delta) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ inside the direct limit of the kernels $\mathrm{Ker}(\delta_\nu)$. By Theorem 3.2, the image of the associated map

$$\rho\colon \mathrm{Ker}(\delta) \longrightarrow H^{2p-1}_{\acute{e}t}(X, \mathbb{Z}_\ell(p))$$

is torsion. After tensoring with $\mathbb{Q}_\ell/\mathbb{Z}_\ell$, that image dies. Therefore the edge map kills the left-hand subgroup and descends uniquely to the quotient. The descended map is precisely Bloch's cycle map. □

The gain of Theorem 4.3 is conceptual rather than formal. It says that the cycle map is not an ad hoc extra arrow appended to Chow groups; it is forced by the fact that the Bloch–Ogus complex already contains the codimension-p divisor calculus, while Merkurjev–Suslin identifies the missing K_2-input needed to compare that complex with Gersten's resolution.

4.5 Top codimension and the Albanese variety

We now specialize to $p = d = \dim X$. At this point the target of Bloch's cycle map admits a concrete algebro-geometric description.

Lemma 4.4. *Let X be smooth projective of dimension d over an algebraically closed field, and let $\ell \neq \mathrm{char}(k)$. Then there are canonical isomorphisms*

$$H^{2d-1}_{\acute{e}t}\big(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)\big) \cong \mathrm{Hom}\big(T_\ell \operatorname{Pic}^0(X), \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)\big) \cong \mathrm{Alb}(X)(k)[\ell^\infty]. \tag{27}$$

Proof. Poincaré duality furnishes a perfect pairing, for every ν,

$$H^{2d-1}_{\acute{e}t}\big(X, \mu_{\ell^\nu}^{\otimes d}\big) \times H^1_{\acute{e}t}(X, \mu_{\ell^\nu}) \longrightarrow H^{2d}_{\acute{e}t}\big(X, \mu_{\ell^\nu}^{\otimes(d+1)}\big) \cong \mu_{\ell^\nu}.$$

Passing to the direct limit in ν gives

$$H^{2d-1}_{\acute{e}t}\big(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)\big) \cong \mathrm{Hom}\big(T_\ell \operatorname{Pic}(X), \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)\big).$$

The Néron–Severi group contributes no divisible ℓ-primary part because $\mathrm{NS}(X)$ is finitely generated, so the connected component is the only part seen after passage to Tate modules:

$$T_\ell \operatorname{Pic}(X) = T_\ell \operatorname{Pic}^0(X).$$

Finally, the duality between $\operatorname{Pic}^0(X)$ and $\mathrm{Alb}(X)$ identifies

$$\mathrm{Hom}\big(T_\ell \operatorname{Pic}^0(X), \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)\big) \cong \mathrm{Alb}(X)(k)[\ell^\infty].$$

This is the desired description. □

Combining Equation (25) and Theorems 4.3 and 4.4, one may rewrite the cohomological side of Bloch's construction in the seminar-friendly form

$$0 \longrightarrow H^{d-1}(X, \mathcal{K}_d) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell \longrightarrow H^{d-1}\big(X, \mathcal{H}^d(\mathbb{Q}_\ell/\mathbb{Z}_\ell(d))\big) \longrightarrow A_0(X)[\ell^\infty] \xrightarrow{\lambda^d_X} \mathrm{Alb}(X)(k)[\ell^\infty]. \tag{28}$$

The point is not that Equation (28) already proves Bloch's theorem. It does not. What it does prove is that, before any Lefschetz-pencil geometry enters, the top-codimension cycle map already lands exactly where one expects: in torsion Albanese. The rest of Bloch's argument, carried out in the remaining sections, shows that this map is in fact an isomorphism.

4.6 The Weil pairing at finite level

It is useful to unravel Theorem 4.4 at the finite level, because this is where the target of λ_X^d stops being an abstract cohomology group and becomes visibly Albanese-theoretic.

Fix $\nu \geq 1$. The Kummer exact sequence

$$0 \longrightarrow \mu_{\ell^\nu} \longrightarrow \mathbf{G}_m \xrightarrow{\ell^\nu} \mathbf{G}_m \longrightarrow 0$$

yields an exact sequence

$$0 \longrightarrow \mathrm{Pic}(X)[\ell^\nu] \longrightarrow H^1_{\acute{e}t}(X, \mu_{\ell^\nu}) \longrightarrow \mathrm{Pic}(X) \xrightarrow{\ell^\nu} \mathrm{Pic}(X),$$

so that

$$H^1_{\acute{e}t}(X, \mu_{\ell^\nu}) \cong \mathrm{Pic}(X)[\ell^\nu].$$

Since $\mathrm{NS}(X)$ is finitely generated, its ℓ^ν-torsion is finite and the connected component controls the divisible part. Thus, for the torsion comparison relevant here, one may replace $\mathrm{Pic}(X)[\ell^\nu]$ by $\mathrm{Pic}^0(X)[\ell^\nu]$.

On the other hand Poincaré duality gives a perfect pairing

$$H^{2d-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes d}) \times H^1_{\acute{e}t}(X, \mu_{\ell^\nu}) \longrightarrow \mu_{\ell^\nu}.$$

After the Kummer identification, this becomes

$$H^{2d-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes d}) \cong \mathrm{Hom}\big(\mathrm{Pic}^0(X)[\ell^\nu], \mu_{\ell^\nu}\big).$$

Finally, the duality between $\mathrm{Pic}^0(X)$ and $\mathrm{Alb}(X)$ produces the finite-level Weil pairing

$$e_{\ell^\nu, X}\colon\ \mathrm{Pic}^0(X)[\ell^\nu] \times \mathrm{Alb}(X)[\ell^\nu] \longrightarrow \mu_{\ell^\nu},$$

which is perfect. Hence there is a canonical isomorphism

$$H^{2d-1}_{\acute{e}t}(X, \mu_{\ell^\nu}^{\otimes d}) \xrightarrow{\sim} \mathrm{Alb}(X)[\ell^\nu], \tag{29}$$

characterized by the fact that a class ξ corresponds to the unique point $a \in \mathrm{Alb}(X)[\ell^\nu]$ such that

$$\langle \xi, \eta \rangle_{\mathrm{PD}} = e_{\ell^\nu, X}(\eta, a) \qquad \text{for every } \eta \in \mathrm{Pic}^0(X)[\ell^\nu].$$

Passing to the direct limit recovers Equation (27). The advantage of Equation (29) is that it makes completely explicit what the cohomology class $\lambda_X^d(z)$ is supposed to measure: it is the unique Albanese torsion point whose Weil pairing against every ℓ^ν-torsion line bundle agrees with the duality pairing defined by the cohomology class of z.

4.7 Curves, Gysin maps, and the Abel–Jacobi meaning of λ_X^d

The preceding discussion still leaves open the concrete geometric meaning of the class $\lambda_X^d(z)$ when z is represented on a curve. In that situation one recovers the usual Abel–Jacobi picture.

Let $i\colon C \hookrightarrow X$ be a smooth connected curve in a smooth projective variety X of dimension d. The pushforward of divisors induces a homomorphism of abelian varieties

$$\psi_i\colon\ \mathrm{Jac}(C) = \mathrm{Pic}^0(C)^\vee \longrightarrow \mathrm{Alb}(X),$$

dual to the restriction map

$$i^*\colon\ \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(C).$$

On the cohomological side, the closed immersion i yields the Gysin map

$$i_*\colon H^1_{ét}(C, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \longrightarrow H^{2d-1}_{ét}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)).$$

For torsion divisor classes on C, Bloch's normalization in codimension one identifies

$$\lambda^1_C\colon \operatorname{Pic}(C)(\ell) \xrightarrow{\sim} H^1_{ét}(C, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1))$$

with the Kummer boundary map.

Proposition 4.5. *Let $z \in A_0(C)[\ell^\infty] \subset \mathrm{CH}^1(C)(\ell)$. Then, after pushing forward to X, one has*

$$\lambda^d_X(i_*z) = i_*(\lambda^1_C(z)) \in H^{2d-1}_{ét}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)).$$

Under the identifications

$$\operatorname{Jac}(C)[\ell^\infty] \cong H^1_{ét}(C, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)), \qquad \operatorname{Alb}(X)[\ell^\infty] \cong H^{2d-1}_{ét}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d)),$$

this is exactly the statement that the square

$$\begin{tikzcd}
\operatorname{Jac}(C)[\ell^\infty] \arrow[r, "\lambda^1_C", "\sim"'] \arrow[d, "\psi_i"'] & H^1_{ét}(C, \mathbb{Q}_\ell/\mathbb{Z}_\ell(1)) \arrow[d, "i_*"] \\
\operatorname{Alb}(X)[\ell^\infty] \arrow[r, "\sim"'] & H^{2d-1}_{ét}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell(d))
\end{tikzcd}$$

commutes.

Proof. At finite level, Bloch's construction starts from the codimension filtration and the divisor map on codimension-$(d-1)$ subvarieties. If a zero-cycle on X is obtained by pushing forward a divisor on C, then every term entering the snake-lemma construction of λ^d_X is obtained from the corresponding codimension-one datum on C by proper pushforward. Compatibility of divisor maps with proper pushforward, together with compatibility of cohomology with supports and Gysin maps, shows that the resulting class in H^{2d-1} is the pushforward of the class constructed on C. This proves the first identity.

For the second assertion, observe that the lower horizontal isomorphism is characterized by the finite-level Weil pairing of Equation (29). Let $a \in \operatorname{Jac}(C)[\ell^\nu]$ and let $\mathcal{M} \in \operatorname{Pic}^0(X)[\ell^\nu]$. Then, by functoriality of Poincaré duality with respect to the Gysin map,

$$\langle i_*\lambda^1_C(a), \mathcal{M}\rangle_X = \langle \lambda^1_C(a), i^*\mathcal{M}\rangle_C.$$

Since λ^1_C is the Kummer map, the right-hand side is exactly the Weil pairing on C:

$$\langle \lambda^1_C(a), i^*\mathcal{M}\rangle_C = e_{\ell^\nu,C}(i^*\mathcal{M}, a).$$

By definition of the dual morphism ψ_i, one has

$$e_{\ell^\nu,C}(i^*\mathcal{M}, a) = e_{\ell^\nu,X}(\mathcal{M}, \psi_i(a)).$$

Hence the class $i_*\lambda^1_C(a)$ corresponds, under Equation (29), to $\psi_i(a)$. Passing to the direct limit proves the commutative square. □

Remark 4.6. Theorem 4.5 is the precise sense in which λ^d_X is an ℓ-adic Abel–Jacobi map for zero-cycles. Whenever a torsion class on X is represented on a smooth curve C, the abstract coniveau class defined in §2 is nothing but the Kummer class of the corresponding point of $\operatorname{Jac}(C)$, transported to X by the Gysin map; under Weil duality, this is exactly its image in $\operatorname{Alb}(X)$. Bloch's theorem says that after all the geometric reductions, this curve-wise interpretation in fact captures every torsion zero-cycle away from the characteristic.

4.8 What remains to prove geometrically

At this point the conceptual picture is sharply defined. The map λ_X^d has been constructed cohomologically, the target has been identified with torsion Albanese via Poincaré duality and the Weil pairing, and for cycles lying on a smooth curve the class λ_X^d is already the expected Abel–Jacobi class. So why is the theorem not finished?

The obstruction is not on the target side but on the source side. A torsion class in $A_0(X)$ need not come presented at the outset by a divisor on a single smooth curve in sufficiently general position. Bloch therefore undertakes a sequence of reductions that manufactures precisely such a presentation, and then studies the remaining ambiguity. After the reductions, one is left with a torsion point of a Jacobian $\mathrm{Jac}(C)$ mapping trivially to $\mathrm{Alb}(X)$. The theorem will follow once one proves that this point already vanishes in the quotient relevant to rational equivalence on X. That is exactly the content of the key lemma proved by Picard–Lefschetz theory in the final sections.

One may therefore read the later geometry as answering a very precise question left open by the cohomological formalism: why is every torsion Albanese class represented by a torsion 0-cycle in a unique way away from the characteristic? The answer is the long geometric chain of reductions that now follows.

5 Surjectivity of the Albanese map on torsion

We first explain the surjectivity part of Theorem 1.1. The key input is the divisibility of $A_0(X)$ over an algebraically closed field.

Proposition 5.1. *For a smooth projective variety X over an algebraically closed field, the group $A_0(X)$ is divisible.*

Idea of proof. Given a degree-zero cycle z, choose an integral curve $C \subset X$ containing the support of z. Since $A_0(C) \cong J(C)(k)$ and $J(C)(k)$ is divisible over an algebraically closed field, z is divisible in $A_0(C)$, hence in $A_0(X)$. □

We may therefore test surjectivity on ℓ-torsion for one prime ℓ at a time.

Proposition 5.2. *For every prime ℓ, the map*

$$A_0(X)[\ell] \longrightarrow \mathrm{Alb}(X)(k)[\ell]$$

is surjective.

Proof. We argue by induction on $d = \dim X$. If $d = 1$, then $A_0(X) = \mathrm{Pic}^0(X)(k)$ and the map is an isomorphism.

Assume $d > 1$, and choose a smooth hyperplane section $i\colon Y \hookrightarrow X$. The functoriality of the Albanese map gives a commutative square

$$\begin{tikzcd} A_0(Y)[\ell] \arrow[r] \arrow[d] & \mathrm{Alb}(Y)(k)[\ell] \arrow[d] \\ A_0(X)[\ell] \arrow[r] & \mathrm{Alb}(X)(k)[\ell]. \end{tikzcd}$$

By induction, the top horizontal map is surjective, so it is enough to prove that

$$\mathrm{Alb}(Y)(k)[\ell] \longrightarrow \mathrm{Alb}(X)(k)[\ell]$$

is surjective.

Ignoring Tate twists, the ℓ-torsion of the Albanese is dual to $H^1_{\acute{e}t}(-, \mathbb{Z}/\ell\mathbb{Z})$. Thus it is enough to prove injectivity of the restriction map

$$H^1_{\acute{e}t}(X, \mathbb{Z}/\ell\mathbb{Z}) \longrightarrow H^1_{\acute{e}t}(Y, \mathbb{Z}/\ell\mathbb{Z}).$$

This is a weak Lefschetz statement for H^1, and in Bloch's presentation it is deduced from Zariski's connectedness theorem for hyperplane sections. Hence the required surjectivity follows. □

Combining Theorems 5.1 and 5.2 yields the surjectivity statement in Theorem 1.1.

6 The injectivity problem and Bloch's sequence of reductions

Fix a prime $\ell \neq \mathrm{char}(k)$ and let

$$z \in A_0(X)[\ell] \qquad \text{with} \qquad \mathrm{alb}_X(z) = 0.$$

We must prove $z \sim_{\mathrm{rat}} 0$. Choose curves $C_i \subset X$ and rational functions $f_i \in k(C_i)^\times$ such that

$$\ell z = \partial(f_1, \ldots, f_n). \tag{30}$$

Set $C := \bigcup_i C_i$.

The argument is a carefully organized simplification of the geometry of (30).

6.1 First reduction: smoothing the support curves

Proposition 6.1 (First reduction). *After replacing X by a blow-up and transporting all data via Theorem 2.1, one may assume:*

(a) each C_i is smooth;

(b) at most two components of $C = \bigcup_i C_i$ pass through any point.

Proof. Blow up successively the singular points of the C_i and the points where three or more components meet. Let $\pi\colon \widetilde{X} \longrightarrow X$ be the resulting morphism, and let $\widetilde{C}_i$ be the strict transforms.

Lift z to a zero-cycle $\widetilde{z}$ on $\widetilde{X}$ with $\pi_*(\widetilde{z}) = z$. Pull each f_i back to $\widetilde{C}_i$ and define

$$w := \ell\widetilde{z} - \widetilde{\partial}(f_1, \ldots, f_n).$$

Then $\pi_*(w) = 0$, so w is supported on the exceptional divisor. Write the exceptional locus as a union of connected components $E_1, \ldots, E_r$ and decompose

$$w = w_1 + \cdots + w_r, \qquad \deg(w_j) = 0, \qquad \mathrm{Supp}(w_j) \subset E_j.$$

Each E_j is a union of projective spaces. Therefore each degree-zero cycle w_j is a sum of principal divisors on lines contained in E_j. Adding these auxiliary lines to the curve configuration replaces the original relation (30) by another one whose supporting curve has the required form. Since blow-up does not change either CH_0 or Alb, the problem is unchanged. □

6.2 Second reduction: cutting down to a surface

Proposition 6.2 (Second reduction)**.** *After replacing X by a smooth projective surface, the problem remains equivalent.*

Proof. Choose a general complete intersection surface

$$Y \subset X$$

of sufficiently high degree containing the curve C. The hypothesis from the first reduction ensures that the singularities imposed by the incidence with C are simple, and for a general choice Y is smooth. By weak Lefschetz, the natural map

$$\mathrm{Alb}(Y) \longrightarrow \mathrm{Alb}(X)$$

is an isomorphism. Since the rational-equivalence relation (30) already takes place on $C \subset Y$, the cycle z belongs to $A_0(Y)$ and is rationally trivial on X if and only if it is rationally trivial on Y. Thus we may replace X by Y. □

From now on, *X is a smooth projective surface.*

6.3 Third reduction: moving the support away from the singular locus

Proposition 6.3 (Third reduction)**.** *One may assume that the support of z consists of smooth points of the curve $C = \bigcup_i C_i$, and that no support point of a summand on C_i lies on $C_i \cap C_j$ for $i \neq j$.*

Proof. Write

$$z = z_1 + \cdots + z_n, \qquad \mathrm{Supp}(z_i) \subset C_i.$$

Because each C_i is smooth, any point of C_i may be moved along a rational equivalence internal to C_i. By moving finitely many points, we arrange that all support points of each z_i lie in $C_i \setminus \bigcup_{j \neq i}(C_i \cap C_j)$ and are smooth on C_i. □

6.4 Fourth reduction: annihilating the class in the generalized Jacobian

At this stage the cycle z determines a class in the generalized Jacobian $J(C)$. Bloch's next goal is not yet to prove $z \sim_{\mathrm{rat}} 0$, but to arrange that the class of ℓz actually vanishes in $J(C)$.

Proposition 6.4 (Fourth reduction)**.** *After modifying the datum inside its rational-equivalence class on X, one may assume that z defines an ℓ-torsion point in $J(C)$, i.e.*

$$[\ell z] = 0 \qquad \textit{in } J(C).$$

Proof. We argue by induction on a combinatorial complexity measure. Let N be the number of pairs (C_i, C_j) such that some point of $C_i \cap C_j$ is a zero or pole of one of the functions f_i, f_j appearing in (30).

If $N = 0$, then the class of ℓz lies in the kernel of

$$J(C) \longrightarrow J(\widetilde{C}),$$

where $\widetilde{C}$ is the normalization of C. This kernel is a connected linear algebraic group, hence divisible on k-points. Therefore, adding a suitable element from this kernel, one may replace z by a rationally equivalent cycle for which $[\ell z] = 0$ in $J(C)$.

Assume now $N > 0$. Renumber so that some zero or pole of f_0 lies on $C_0 \cap C_\infty$. By adding auxiliary curves in general position, one may arrange that C_0 and C_∞ become linearly equivalent very ample divisors and that a general member of the corresponding pencil is smooth. Write

$$z = z_0 - z_\infty, \tag{31}$$

with $\mathrm{Supp}(z_0) \subset C_0$, $\mathrm{Supp}(z_\infty) \subset C_\infty$, both of degree 0, and such that the class of ℓz_0 in $J(C_0)$ is represented by a degree-zero divisor δ supported on $C_0 \cap C_\infty$.

Blow up the base locus of the pencil to obtain

$$f\colon \mathcal{C} \longrightarrow \mathbb{P}^1.$$

The divisor δ determines a section

$$\Delta\colon \mathbb{P}^1 \longrightarrow \mathrm{Pic}(\mathcal{C}/\mathbb{P}^1).$$

Consider the inverse image under multiplication by ℓ,

$$\ell^{-1}(\Delta) \subset \mathrm{Pic}(\mathcal{C}/\mathbb{P}^1).$$

This is a principal homogeneous space under the finite flat group scheme $\mathrm{Pic}(\mathcal{C}/\mathbb{P}^1)[\ell]$, and it is non-empty because it contains the point corresponding to z_0 over $0 \in \mathbb{P}^1$. Let Z be the normalization of an irreducible component dominating the base. A crucial lemma, proved below, shows that this component maps onto a non-empty open subset of $\mathbb{P}^1$.

Choose points $w_0, w_\infty \in Z$ lying over 0 and ∞. The universal property of the relative Picard provides, after pullback to Z, a divisor D on $Z \times_{\mathbb{P}^1} \mathcal{C}$ such that

$$\ell D \sim \pi^*\Delta + p^*(t) \tag{32}$$

for some divisor t on Z, where π and p are the projections in the diagram

$$\begin{tikzcd}
Z \times_{\mathbb{P}^1} \mathcal{C} \arrow[r, "\pi"] \arrow[d, "p"'] & \mathcal{C} \arrow[r, "\sigma"] \arrow[d, "f"] & X \\
Z \arrow[r, "q"] & \mathbb{P}^1. &
\end{tikzcd}$$

Fix a divisor w on Z with

$$\ell w \sim (w_0) - (w_\infty).$$

Intersecting $p^*(w)$ with D and pushing forward to X, Bloch produces a cycle z'_∞ supported on C_∞ such that

$$[z'_\infty] = [\delta] \quad \text{in } J(C_\infty).$$

Hence there exist functions g_i on the C_i with no zeros or poles at the nodes and such that replacing (f_i) by $(f_i g_i)$ decreases the complexity N. Induction on N proves the result. □

Remark 6.5. This step is the technical heart of the moving argument. The passage to the relative Picard torsor is what allows one to transport an ℓ-divisibility statement from one fibre of the pencil to another while controlling the support of the cycle.

7 The relative Picard torsor and the auxiliary geometric lemma

The construction in the fourth reduction rests on the following geometric fact.

Lemma 7.1 (Bloch's auxiliary lemma)**.** *With notation as above, every irreducible component of the finite group scheme*

$$\mathrm{Pic}(\mathcal{C}/\mathbb{P}^1)[\ell] \longrightarrow \mathbb{P}^1$$

maps onto a non-empty open subset of $\mathbb{P}^1$*.*

Proof. Fix a smooth fibre C_t of the pencil. The fibre of $\mathrm{Pic}(\mathcal{C}/\mathbb{P}^1)[\ell]$ at t is, up to the usual Tate twist, the finite group $H^1_{\acute{e}t}(C_t, \mathbb{Z}/\ell\mathbb{Z})$. The point is therefore to show that an ℓ-torsion line bundle on one smooth fibre extends étale-locally in the base.

Now the family $f\colon \mathcal{C} \longrightarrow \mathbb{P}^1$ is smooth away from the finite discriminant set $\Sigma \subset \mathbb{P}^1$. Over the strict henselization of the local ring at a point $t \notin \Sigma$, every finite étale cover of the fibre lifts uniquely to a finite étale cover of the total space. Equivalently, every ℓ-torsion point in the fibre of the relative Picard spreads out over some étale neighbourhood of t. Thus each closed point of a fibre belongs to a component whose image contains a non-empty open neighbourhood of t, which proves the lemma. □

8 Fifth reduction: moving to a smooth curve in the pencil

After the fourth reduction we have a very ample curve $C \subset X$, not necessarily smooth, satisfying

$$\mathrm{Supp}(z) \subset C_{\mathrm{sm}}, \qquad [\ell z] = 0 \text{ in } J(C), \qquad \mathrm{alb}_X(z) = 0.$$

Bloch next shows that one may replace C by a smooth member of a pencil.

Proposition 8.1 (Fifth reduction)**.** *The problem reduces to the case where $C \subset X$ is a smooth hyperplane section and the class of z in $J(C) = \mathrm{Jac}(C)$ is ℓ-torsion and lies in the kernel of*

$$\mathrm{Jac}(C) \longrightarrow \mathrm{Alb}(X).$$

Proof. Take a Lefschetz pencil with special fibre $C = C_0$, and let Z be the normalization of an irreducible component of $\mathrm{Pic}(\mathcal{C}/\mathbb{P}^1)[\ell]$ through the point representing $[z] \in J(C_0)$. By Theorem 7.1 we may choose a point $w_1 \in Z$ lying over a general parameter $1 \in \mathbb{P}^1$ for which the fibre C_1 is smooth.

Exactly as in the fourth reduction, there is a divisor D on $Z \times_{\mathbb{P}^1} \mathcal{C}$ such that, up to rational equivalence and vertical components,

$$\ell D \sim p^*(t)$$

for some divisor t on Z. Let w be a divisor on Z satisfying

$$\ell w \sim (w_1) - (w_0).$$

Then the cycle

$$z' := \pi_*\big(p^*(w_1) \cdot D\big)$$

is rationally equivalent to z, is supported on the smooth fibre C_1, and satisfies $[\ell z'] = 0$ in $\mathrm{Jac}(C_1)$. Replacing (X, C, z) by (X, C_1, z') proves the claim. □

We are therefore reduced to the following situation:

> X is a smooth projective surface, $C \subset X$ is a smooth hyperplane section, $z \in A_0(C)$ is of ℓ-power torsion in $\mathrm{Jac}(C)$, and its image in $\mathrm{Alb}(X)$ is zero.

9 The geometry of the map $\mathrm{Jac}(C) \longrightarrow \mathrm{Alb}(X)$

Before turning to the key lemma, it is useful to isolate the geometry hidden in the map from the Jacobian of the hyperplane section to the Albanese of the ambient surface. This is the place where Bloch's argument ceases to be a mere moving lemma for cycles and becomes a statement about the vanishing part of a Lefschetz pencil.

Let X be a smooth projective surface and let $i\colon C \hookrightarrow X$ be a smooth connected hyperplane section. The pushforward of degree-zero divisors defines a homomorphism

$$\phi_C\colon \mathrm{Jac}(C) = A_0(C) \longrightarrow A_0(X) \xrightarrow{\mathrm{alb}_X} \mathrm{Alb}(X).$$

Dually, one has the restriction map of Picard varieties

$$i^*\colon \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(C) = \mathrm{Jac}(C)^\vee.$$

Since C is ample, the restriction on tangent spaces

$$H^1(X, \mathcal{O}_X) \longrightarrow H^1(C, \mathcal{O}_C)$$

is injective; equivalently, i^* is an immersion of abelian varieties. Dualizing, one sees that ϕ_C is surjective with connected kernel.

Proposition 9.1. *Let X be a smooth projective surface and let $C \subset X$ be a smooth hyperplane section. Then there is an exact sequence of abelian varieties*

$$0 \longrightarrow K(C/X) \longrightarrow \mathrm{Jac}(C) \xrightarrow{\phi_C} \mathrm{Alb}(X) \longrightarrow 0,$$

where $K(C/X)$ is a connected abelian subvariety of $\mathrm{Jac}(C)$.

Proof. The morphism ϕ_C is dual to the restriction map $i^*\colon \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(C)$. For a smooth ample divisor on a surface, weak Lefschetz in degree 1 implies that the induced map

$$H^1_{\acute{e}t}(X, \mathbb{Z}/\ell\mathbb{Z}) \longrightarrow H^1_{\acute{e}t}(C, \mathbb{Z}/\ell\mathbb{Z})$$

is injective for every prime ℓ, hence i^* has finite kernel. But i^* also induces an injection on tangent spaces, so the kernel is trivial and i^* is a closed immersion of abelian varieties. Dualizing gives a surjection $\phi_C\colon \mathrm{Jac}(C) \twoheadrightarrow \mathrm{Alb}(X)$ with connected kernel. □

The resulting geometry may be summarized by the dual diagram

$$\begin{tikzcd}
0 \arrow[r] & K(C/X) \arrow[r] & \mathrm{Jac}(C) \arrow[r, "\phi_C"] & \mathrm{Alb}(X) \arrow[r] & 0 \\
0 \arrow[r] & \mathrm{Pic}^0(X) \arrow[r, "i^*"] & \mathrm{Pic}^0(C) \arrow[r] & K(C/X)^\vee \arrow[r] & 0
\end{tikzcd}$$

where the second row is the dual exact sequence.

The point of Bloch's key lemma is that the ℓ-primary torsion of $K(C/X)$ can be recognized in two a priori different ways:

(i) as the kernel of $\mathrm{Jac}(C)[\ell^\infty] \longrightarrow \mathrm{Alb}(X)[\ell^\infty]$;

(ii) as the subgroup of $\mathrm{Jac}(C)[\ell^\infty]$ represented by zero-cycles on C that become rationally equivalent to zero on X.

What makes these two kernels coincide is precisely the fact that $K(C/X)$ is governed by vanishing cohomology.

Proposition 9.2. *Let $\ell \neq \mathrm{char}(k)$. With notation as above, the ℓ-adic Tate module of $K(C/X)$ identifies with the kernel of the Gysin map*

$$H^1_{\acute{e}t}(C, \mathbb{Z}_\ell) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}_\ell).$$

Equivalently, after passing to $\mathbb{Q}_\ell/\mathbb{Z}_\ell$-coefficients, one has

$$K(C/X)[\ell^\infty] \cong \mathrm{Ker}\Big(H^1_{\acute{e}t}(C, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)\Big).$$

Proof. The map ϕ_C is dual to $i^*\colon \mathrm{Pic}^0(X) \longrightarrow \mathrm{Pic}^0(C)$. On ℓ-adic Tate modules, duality for abelian varieties identifies

$$T_\ell \,\mathrm{Jac}(C) \cong H^1_{\acute{e}t}(C, \mathbb{Z}_\ell)(1), \qquad T_\ell \,\mathrm{Alb}(X) \cong H^1_{\acute{e}t}(X, \mathbb{Z}_\ell)(1)^\vee.$$

Under Poincaré duality on the surface X, the dual of the restriction map

$$H^1_{\acute{e}t}(X, \mathbb{Z}_\ell) \longrightarrow H^1_{\acute{e}t}(C, \mathbb{Z}_\ell)$$

is the Gysin map

$$H^1_{\acute{e}t}(C, \mathbb{Z}_\ell)(-1) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}_\ell).$$

Therefore the kernel of $T_\ell \,\mathrm{Jac}(C) \longrightarrow T_\ell \,\mathrm{Alb}(X)$ identifies, up to the harmless Tate twist suppressed throughout Bloch's argument, with the kernel of the Gysin map. Since $K(C/X)$ is the connected kernel of ϕ_C, its Tate module is exactly this kernel, and passing to the direct limit over ℓ^ν yields the second statement. □

This is the conceptual bridge between the algebro-geometric part of the proof and the monodromy part. Once a Lefschetz pencil through C is chosen, the kernel $K(C/X)$ is no longer an abstract abelian subvariety: it becomes the vanishing part of the local system $R^1 f_* \mathbb{Z}_\ell$ on the smooth locus of the pencil, hence a space generated by vanishing cycles and controlled by Picard–Lefschetz reflections.

Remark 9.3. From this viewpoint, Bloch's key lemma says that the *vanishing abelian subvariety* $K(C/X)$ does not contain any mysterious prime-to-the-characteristic torsion beyond what already comes from rational equivalences produced by the universal divisor on the relative Picard torsor. The monodromy argument is precisely what turns this geometric intuition into an actual proof.

9.1 The vanishing abelian scheme over the smooth locus

Let $f\colon \widetilde{X} \longrightarrow \mathbb{P}^1$ be the blow-up of a Lefschetz pencil on X and let $U \subset \mathbb{P}^1$ be the open set over which the fibres are smooth. Over U, the relative Picard functor

$$J_U := \mathrm{Pic}^0_{\widetilde{X}_U/U}$$

is an abelian scheme whose fibre over $u \in U$ is $\mathrm{Jac}(C_u)$. The Albanese map on each smooth fibre assembles into a morphism of abelian schemes

$$h_U\colon J_U \longrightarrow \mathrm{Alb}(X) \times U.$$

Because each fibre map $\mathrm{Jac}(C_u) \longrightarrow \mathrm{Alb}(X)$ is surjective with connected kernel, the scheme-theoretic kernel

$$K_U := \mathrm{Ker}(h_U)$$

is itself an abelian subscheme of J_U over U. Fibrewise, $K_{U,u} = K(C_u/X)$ is exactly the connected kernel introduced above. Thus the single abelian variety $K(C/X)$ appearing in the statement of the key lemma is best thought of as one fibre of a whole family of "vanishing abelian varieties" varying along the pencil.

This family-theoretic viewpoint is the cleanest bridge between the geometry of Jacobians and the cohomological formalism of vanishing cycles. Indeed, the ℓ-adic Tate module of J_U is the lisse sheaf

$$T_\ell(J_U) \cong R^1 f_{U,*}\mathbb{Z}_\ell(1),$$

while the constant abelian scheme $\mathrm{Alb}(X) \times U$ has Tate module the constant sheaf

$$T_\ell(\mathrm{Alb}(X) \times U) \cong H^1_{\acute{e}t}(X, \mathbb{Z}_\ell)(1)^\vee \otimes \underline{\mathbb{Z}_\ell}_U.$$

Via Poincaré duality on X, the morphism on Tate modules induced by h_U is dual to the fibrewise restriction map

$$H^1_{\acute{e}t}(X, \mathbb{Z}_\ell) \otimes \underline{\mathbb{Z}_\ell}_U \longrightarrow R^1 f_{U,*}\mathbb{Z}_\ell.$$

Consequently the Tate module of K_U is the kernel of the dual map, hence the same object as the vanishing local system up to the customary Tate twist.

Proposition 9.4. *With notation as above, there is a short exact sequence of lisse $\mathbb{Z}_\ell$-sheaves on U*

$$0 \longrightarrow H^1_{\acute{e}t}(X, \mathbb{Z}_\ell) \otimes \underline{\mathbb{Z}_\ell}_U \longrightarrow R^1 f_{U,*}\mathbb{Z}_\ell \longrightarrow \mathbb{V}_{\mathrm{van}} \longrightarrow 0,$$

where $\mathbb{V}_{\mathrm{van}}$ is the vanishing local system of the pencil. Moreover,

$$T_\ell(K_U) \cong \mathbb{V}_{\mathrm{van}}(1)$$

as lisse sheaves on U.

Proof. For each geometric point $\bar{u} \longrightarrow U$, weak Lefschetz gives an injection

$$H^1_{\acute{e}t}(X, \mathbb{Z}_\ell) \hookrightarrow H^1_{\acute{e}t}(C_{\bar{u}}, \mathbb{Z}_\ell).$$

These injections are compatible with specialization and therefore globalize to an injection of lisse sheaves on U. The cokernel is, by definition, the quotient of fibrewise cohomology by the part coming from the ambient surface; this is exactly the vanishing local system $\mathbb{V}_{\mathrm{van}}$. On the abelian-scheme side, the exact sequence

$$0 \longrightarrow K_U \longrightarrow J_U \xrightarrow{h_U} \mathrm{Alb}(X) \times U \longrightarrow 0$$

yields after applying $T_\ell(-)$ a short exact sequence of lisse sheaves. Under the identifications above, this sequence is dual to the one just written. Poincaré duality identifies the dual quotient with $\mathbb{V}_{\mathrm{van}}(1)$, giving the claim. □

The moral is worth stating plainly: the kernel $K(C/X)$ is not merely "the kernel of a map of abelian varieties". Inside a Lefschetz pencil it is the abelian avatar of the new cohomology created by the hyperplane section. The monodromy representation on $\mathbb{V}_{\mathrm{van}}$ is therefore exactly the monodromy representation on the Tate modules of the fibres of K_U.

This may be summarized by the commutative picture

$$\begin{tikzcd}
0 \arrow[r] & T_\ell(K_U) \arrow[r] \arrow[d, "\sim"'] & T_\ell(J_U) \arrow[r] \arrow[d, "\sim"] & T_\ell(\mathrm{Alb}(X) \times U) \arrow[r] \arrow[d, "\sim"] & 0 \\
0 \arrow[r] & \mathbb{V}_{\mathrm{van}}(1) \arrow[r] & R^1 f_{U,*}\mathbb{Z}_\ell(1) \arrow[r] & H^1_{\acute{e}t}(X, \mathbb{Z}_\ell)(1)^\vee \otimes \underline{\mathbb{Z}_\ell}_U \arrow[r] & 0
\end{tikzcd}$$

where the lower row is dual, via Poincaré duality, to the restriction sequence above. This is the precise sense in which the abelian scheme K_U is the "vanishing abelian scheme" attached to the pencil.

9.2 Nearby cycles, variation, and the Picard–Lefschetz transvection

The previous description globalizes the smooth part of the story; the singular fibres explain how monodromy acts on it. Fix a critical value $s \in \mathbb{P}^1 \setminus U$, choose a geometric generic point $\bar{\eta}_s$ of the strict henselization at s, and write $C_{\bar{\eta}_s}$ for the nearby smooth fibre. The nearby-cycle triangle

$$i_s^* R f_* \mathbb{Z}_\ell \longrightarrow R\Psi_s \mathbb{Z}_\ell \longrightarrow R\Phi_s \mathbb{Z}_\ell \xrightarrow{+1}$$

packages the defect of smooth base change at s. In the Lefschetz situation relevant here, the singular fibre has exactly one ordinary double point, so $R\Phi_s\mathbb{Z}_\ell$ is concentrated in degree 1 and

$$H^1(R\Phi_s\mathbb{Z}_\ell) \cong \mathbb{Z}_\ell \cdot \delta_s$$

is free of rank one, generated by a vanishing cycle δ_s defined up to sign.

The variation morphism

$$\mathrm{var}_s \colon H^1(R\Phi_s\mathbb{Z}_\ell) \longrightarrow H^1(C_{\bar{\eta}_s}, \mathbb{Z}_\ell)$$

identifies its image with the line spanned by δ_s. The local monodromy operator T_s satisfies $N_s := T_s - 1$ of rank one, and one has the Picard–Lefschetz formula

$$T_s(x) = x \pm \langle x, \delta_s \rangle \, \delta_s.$$

Thus each singular fibre contributes exactly one transvection on the vanishing lattice. Because the various δ_s generate $\mathbb{V}_{\mathrm{van}}$ and are conjugate under the geometric monodromy, the entire monodromy of the vanishing abelian scheme is generated by these rank-one reflections.

For the proof of Bloch's key lemma, this is the decisive geometric input. The kernel component $Z \subset V[\ell^\nu]$ chosen later is not arbitrary: it is forced on us by the fact that a single nontrivial point in the vanishing torsion, once moved around the discriminant by Picard–Lefschetz transvections, spreads across the full vanishing part. That is precisely why Bloch can rewrite an arbitrary class in the fibrewise kernel as a zero-sum combination of points lying on one irreducible component.

9.3 Lefschetz pencils, vanishing cycles, and precise references

The monodromy input used from this point onward is the standard ℓ-adic Picard–Lefschetz package for a Lefschetz pencil on a smooth projective surface. Choose a very ample line

bundle on X, choose a pencil through the fixed smooth member C, and blow up the base locus. One obtains a morphism

$$f\colon \widetilde{X} \longrightarrow \mathbb{P}^1$$

whose smooth fibres are smooth hyperplane sections and whose singular fibres have exactly one ordinary double point. If $U \subset \mathbb{P}^1$ denotes the complement of the discriminant and $j\colon U \hookrightarrow \mathbb{P}^1$ the open immersion, then $R^1 f_{U,*}\mathbb{Z}_\ell$ is a lisse $\mathbb{Z}_\ell$-sheaf. For every singular value $s \in \mathbb{P}^1 \setminus U$, with inclusion $i_s\colon \{s\} \hookrightarrow \mathbb{P}^1$, the nearby- and vanishing-cycle formalism furnishes an exact triangle

$$i_s^* R f_* \mathbb{Z}_\ell \longrightarrow R\Psi_s \mathbb{Z}_\ell \longrightarrow R\Phi_s \mathbb{Z}_\ell \xrightarrow{+1},$$

and, because the singularity is an ordinary quadratic singularity on a family of curves, the complex $R\Phi_s\mathbb{Z}_\ell$ is concentrated in degree 1 and its unique nonzero stalk is a free rank-one $\mathbb{Z}_\ell$-module generated by a vanishing cycle

$$\delta_s \in H^1\big(C_{\bar{\eta}_s}, \mathbb{Z}_\ell\big),$$

well-defined up to sign. The geometric local monodromy around s is unipotent of index 2 in this situation and acts by the Picard–Lefschetz reflection

$$\sigma_s(x) = x \pm \langle x, \delta_s \rangle\, \delta_s.$$

These are exactly the facts used by Bloch in the proof of the key lemma.

For the original sources, one may separate the local and global inputs as follows. The local vanishing-cycle formalism and the Picard–Lefschetz formula are developed in SGA 7 II, especially Exp. XIII and Exp. XV; the global monodromy consequences for Lefschetz pencils satisfying condition (A) are organized in Exp. XVIII. For the way these statements are inserted into the present zero-cycle argument, see Bloch [1, § 4, esp. pp. 121–126]. A modern companion reference is Illusie [3], which is useful for keeping clear which part of the argument is local vanishing-cycle theory and which part is global monodromy for the pencil.

More precisely, the nearby- and vanishing-cycle formalism for one-parameter degenerations is laid out in SGA 7 II, Exp. XIII, while the Picard–Lefschetz description of the local monodromy in the ordinary quadratic case is treated in Exp. XV; the existence and irreducibility statements for Lefschetz pencils satisfying condition (A) are organized in Exp. XVIII. Bloch applies exactly this package in [1, § 4, esp. pp. 121–126]. For a modern roadmap through the same circle of ideas, see Illusie [3, §§ 3–5].

Etale-locally near a nodal point, the family is modeled by

$$xy = t$$

over a strict henselian trait with parameter t. The special fiber has one ordinary double point, the generic fiber is smooth, and the vanishing-cycle complex is concentrated in degree 1 with rank-one stalk. The variation map identifies its image inside H^1 of the nearby smooth fiber with the line generated by the vanishing cycle δ_s. In this model the local monodromy is a transvection, and the Picard–Lefschetz formula says exactly that the transvection coefficient is computed by the cup-product pairing with δ_s.

Proposition 9.5 (Lefschetz package for the proof)**.** *With notation as above, let C_u be a smooth fibre of the pencil. Then:*

(i) the vanishing submodule $V_{\mathrm{van}} \subset H^1_{\acute{e}t}(C_u, \mathbb{Z}_\ell)$ generated by the vanishing cycles is the kernel of the Gysin map

$$H^1_{\acute{e}t}(C_u, \mathbb{Z}_\ell) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}_\ell);$$

(ii) the vanishing cycles form one monodromy orbit up to sign and generate V_{van} as a $\mathbb{Z}_\ell$-module;

(iii) the local monodromy transformations around the singular fibres act on V_{van} by Picard–Lefschetz reflections.

Explanation of the input. For statement (i), the Leray spectral sequence for f together with weak Lefschetz identifies the monodromy-invariant part of $H^1(C_u, \mathbb{Z}_\ell)$ with the image of $H^1(X, \mathbb{Z}_\ell)$. Dually, the quotient by that invariant part is the kernel of the Gysin map

$$H^1_{\acute{e}t}(C_u, \mathbb{Z}_\ell) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}_\ell),$$

so the vanishing submodule is exactly the part of $H^1(C_u)$ that is new with respect to the ambient surface. This is the cohomological incarnation of the abelian subvariety $K(C/X) \subset \mathrm{Jac}(C)$ introduced earlier. After quotienting by the monodromy-invariant part, one may summarize the situation by the exact sequence

$$0 \longrightarrow H^1_{\acute{e}t}(X, \mathbb{Z}_\ell) \longrightarrow H^1_{\acute{e}t}(C_u, \mathbb{Z}_\ell) \longrightarrow V_{\mathrm{van}} \longrightarrow 0,$$

which is the cohomological shadow of the quotient of $\mathrm{Jac}(C_u)$ by the image of $\mathrm{Pic}^0(X)$.

For statements (ii) and (iii), one uses the local description of an ordinary quadratic singularity. The vanishing-cycle complex has rank one at each critical value, hence each singular fibre contributes a single class δ_s, and the local monodromy acts by adding to any class the appropriate multiple of δ_s dictated by the cup-product pairing. Global irreducibility of the vanishing part under the geometric monodromy implies that the various δ_s lie in one orbit up to sign and generate the full vanishing submodule. No finer form of the theorem is needed below: Bloch only uses that the vanishing part is generated by these cycles and that Picard–Lefschetz reflections move one vanishing cycle against the others in a controlled, intersection-theoretic way. What matters for the key lemma is that the monodromy group does not preserve an unexpected proper submodule of the vanishing lattice compatible with the pairing; once a nonzero torsion class is transported around the pencil, repeated reflections force it to spread through the whole vanishing part. □

10 The key lemma: equality of kernels

The decisive input is the statement Bloch calls the key lemma.

Lemma 10.1 (Key lemma)**.** *Let X be a smooth projective surface, let $C \subset X$ be a smooth hyperplane section, and let $\ell \neq \mathrm{char}(k)$. Then the two maps*

$$\mathrm{Jac}(C)[\ell^\infty] \longrightarrow \mathrm{Alb}(X)(k)[\ell^\infty], \qquad \mathrm{Jac}(C)[\ell^\infty] \longrightarrow A_0(X)[\ell^\infty]$$

have the same kernel.

Assuming this for a moment, the proof of injectivity in Theorem 1.1 is immediate: if $z \in A_0(C)$ is ℓ-primary torsion and maps to zero in $\mathrm{Alb}(X)$, then its class in $\mathrm{Jac}(C)$ lies in the kernel of the first map, hence by the key lemma also in the kernel of the second, so $z \sim_{\mathrm{rat}} 0$ in X.

10.1 The pencil and the kernel scheme

Take a Lefschetz pencil with $C = C_0$ and total space

$$f\colon \mathcal{C} \longrightarrow \mathbb{P}^1.$$

There is a natural morphism of group schemes over the base

$$h\colon\ \mathrm{Pic}(\mathcal{C}/\mathbb{P}^1) \longrightarrow \mathrm{Alb}(X) \times \mathbb{P}^1,$$

obtained by composing the Abel map on each fibre with the pushforward to the Albanese of X. Let

$$V := \mathrm{Ker}(h), \qquad V[\ell^\nu] \subset V$$

be the ℓ^ν-torsion subgroup schemes.

Bloch compares this scheme-theoretic kernel with vanishing cohomology. On the fibre over a smooth parameter t one has, up to twist,

$$\mathrm{Pic}(C_t)[\ell^\nu] \cong H^1_{\acute{e}t}(C_t, \mathbb{Z}/\ell^\nu\mathbb{Z}),$$

and the map to $\mathrm{Alb}(X)[\ell^\nu]$ corresponds to the Gysin map

$$H^1_{\acute{e}t}(C_t, \mathbb{Z}/\ell^\nu\mathbb{Z}) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}/\ell^\nu\mathbb{Z}).$$

Thus the fibrewise kernel is

$$V'_t := \mathrm{Ker}\Big(H^1_{\acute{e}t}(C_t, \mathbb{Z}/\ell^\nu\mathbb{Z}) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}/\ell^\nu\mathbb{Z})\Big). \tag{33}$$

Passing to the direct limit over ν gives

$$\mathrm{Ker}\Big(H^1_{\acute{e}t}(C_t, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell)\Big) \cong V_t[\ell^\infty].$$

10.2 Vanishing cycles

Fix a geometric base point $0 \in \mathbb{P}^1 \setminus \Sigma$, where Σ is the discriminant locus of the pencil. Let

$$V' := \mathrm{Ker}\Big(H^1_{\acute{e}t}(C, \mathbb{Z}_\ell) \longrightarrow H^3_{\acute{e}t}(X, \mathbb{Z}_\ell)\Big).$$

By Theorem 9.5, the subgroup V' is the vanishing cohomology of the pencil. It is generated by the vanishing cycles δ_i, one for each singular fibre, and these cycles are conjugate under the monodromy representation of

$$\pi_1(\mathbb{P}^1 \setminus \Sigma, 0).$$

If σ_i denotes the Picard–Lefschetz transformation associated with δ_i, then for any $\chi \in V'$ one has

$$\sigma_i(\chi) = \chi \pm \langle \chi, \delta_i \rangle \delta_i, \tag{34}$$

where $\langle\ ,\ \rangle$ is the intersection pairing on $H^1(C, \mathbb{Z}_\ell)$.

10.3 A claim on the geometry of $V[\ell^\nu]$

The crucial intermediate statement is the following.

Proposition 10.2 (Geometric claim inside the ℓ^ν-torsion kernel)**.** *There is an irreducible component $Z \subset V[\ell^\nu]$ such that every element*

$$\gamma' \in V_0'[\ell^\nu]$$

can be written in the form

$$\gamma' = \sum_i n_i \gamma_i, \qquad \gamma_i \in Z_0, \qquad \sum_i n_i = 0.$$

Proof. Let Z be the irreducible component containing one vanishing cycle δ_1; since all vanishing cycles are monodromy-conjugate, Z contains them all. As the δ_i generate the vanishing cohomology modulo ℓ^ν, any class γ' can be written

$$\gamma' = \sum_i m_i \delta_i.$$

The issue is to arrange the extra relation $\sum_i m_i \equiv 0 \pmod{\ell^\nu}$.

Suppose $m := \sum_i m_i$ and choose an index 1 such that m_1 divides all m_i after relabelling. Bloch's trick is to select a Picard–Lefschetz transformation σ_q for which the scalar $\langle \delta_1, \delta_q \rangle / \delta_q$ is invertible modulo ℓ^ν. If no such q existed, then each difference $\sigma_q(\delta_1) - \delta_1$ would lie in $\ell V'$, and because the $\sigma_q(\delta_1)$ generate the vanishing cohomology this would force an impossible divisibility property on V'. In particular one contradicts the fact, recalled by Bloch, that $\mathrm{rk}(V')$ is even.

Choose r so that repeated application of σ_q adds precisely m/m_1 copies of δ_q to δ_1 modulo ℓ^ν. Then

$$\sigma_q^r(\delta_1) \in Z_0$$

and replacing the term $m_1\delta_1$ by $m_1\sigma_q^r(\delta_1)$ changes the total coefficient sum by $-m$. Thus one rewrites γ' as a linear combination of points in Z_0 with total coefficient sum 0, exactly as required. □

10.4 Completion of the proof of the key lemma

We now finish the proof of Theorem 10.1.

Proof of Theorem 10.1. Let $\gamma \in \mathrm{Jac}(C)[\ell^\nu]$ map to zero in $\mathrm{Alb}(X)$. Its image in the fibre of $V[\ell^\nu]$ over 0 corresponds to an element

$$\gamma' \in V_0'[\ell^\nu].$$

By Theorem 10.2, after multiplying by a suitable power of ℓ if necessary and passing to the corresponding torsor as in the fourth and fifth reductions, we may write

$$\gamma' = \sum_i n_i \gamma_i, \qquad \gamma_i \in Z_0, \qquad \sum_i n_i = 0.$$

Pull back the universal divisor along the component Z. Exactly as before, one obtains a divisor D on $Z \times_{\mathbb{P}^1} \mathcal{C}$ and, for the zero-cycle $\sum_i n_i(\gamma_i)$ on Z, a class in $A_0(X)$ represented by

$$\pi_*\Big(p^*\Big(\sum_i n_i \gamma_i\Big) \cdot D\Big).$$

Because the coefficients sum to zero, the corresponding class belongs to the Jacobian $\mathrm{Jac}(Z)$, which is divisible over k. Therefore the resulting zero-cycle on X is rationally equivalent to zero. Tracing the construction back shows precisely that the original γ maps to zero in $A_0(X)$. Thus the kernel of

$$\mathrm{Jac}(C)[\ell^\infty] \longrightarrow \mathrm{Alb}(X)(k)[\ell^\infty]$$

is contained in the kernel of

$$\mathrm{Jac}(C)[\ell^\infty] \longrightarrow A_0(X)[\ell^\infty].$$

The reverse inclusion is immediate from the factorization

$$\mathrm{Jac}(C)[\ell^\infty] \longrightarrow A_0(X)[\ell^\infty] \xrightarrow{\mathrm{alb}_X} \mathrm{Alb}(X)(k)[\ell^\infty].$$

Hence the two kernels coincide. □

11 Conclusion of Bloch's theorem

We can now assemble the argument.

Proof of Theorem 1.1. The surjectivity of the Albanese map on torsion was proved in Theorem 5.2 together with the divisibility of $A_0(X)$.

Fix a prime $\ell \neq \mathrm{char}(k)$. Let $z \in A_0(X)[\ell^\infty]$ satisfy $\mathrm{alb}_X(z) = 0$. By the five reductions of Theorems 6.1 to 6.4 and 8.1, we may assume that X is a smooth projective surface, that z is supported on a smooth hyperplane section $C \subset X$, and that the class of z defines an ℓ-primary torsion point of $\mathrm{Jac}(C)$ mapping to zero in $\mathrm{Alb}(X)$. By the key lemma Theorem 10.1, this class lies in the kernel of

$$\mathrm{Jac}(C)[\ell^\infty] \longrightarrow A_0(X)[\ell^\infty],$$

so $z \sim_{\mathrm{rat}} 0$ in X. This proves injectivity on the ℓ-primary torsion subgroup.

Since this holds for every $\ell \neq \mathrm{char}(k)$, the Albanese map induces an isomorphism away from the characteristic. Together with surjectivity on the whole torsion subgroup, this is exactly the statement proved by Bloch. □

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